

Group Theory - Lecture 7

Summary of last week's lecture

Lie group		Lie algebra
i) G	$\mapsto$	$\mathfrak{g} \cong T_e G$
ii) $\phi: G \rightarrow H$	$\mapsto$	$\varphi = d\phi_e: \mathfrak{g} \rightarrow \mathfrak{h}$
iii) $H \subset G$	$\mapsto$	$\mathfrak{h} \subset \mathfrak{g}$
iv) $\exists!$ simply connected	$\longleftrightarrow$	$\mathfrak{g}$

Lie group $\tilde{G}$ other: $G = \tilde{G}/\Gamma, \Gamma \subset Z(\tilde{G})$

v) unique connected $H \subset G$	$\mapsto$	$\mathfrak{h} \subset \mathfrak{g}$
vi) $\exists! \phi: G \rightarrow H, G$ simply connected, H connected		$\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$

Representations

$\phi: G \rightarrow GL(V)$	$\mapsto$	$\varphi = d\phi_e: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$
$\exists! \phi: \tilde{G} \rightarrow GL(V)$	$\longleftrightarrow$	$\varphi: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$
$\uparrow$		
simply conn Lie group with		
Lie algebra $\mathfrak{g}, \tilde{G} = \tilde{G}/\Gamma, \ker(\phi) \cap \Gamma$		

Exponential map

G : Lie group, $\text{Lie}(G) = \mathfrak{g}$. For every $v \in \mathfrak{g}$ can define subalgebra $\mathfrak{h} = \text{span}(v) \subset \mathfrak{g}$. Using v from above table $\rightarrow$ 1-parameter subgroup $H \subset G$ (unique connected)

$$\phi_v: (\mathbb{R}, +) \rightarrow G \quad \varphi_v: \mathfrak{h} \rightarrow \mathfrak{g}, v \mapsto v$$

$$\phi_v(s) \phi_v(t) = \phi_v(s+t)$$

$$(d\phi_v)_e = v$$

i) $\phi_v(t): \mathbb{R} \rightarrow G$ (integral curve of X_v)

$$\dot{\gamma}(t) = X_v(\gamma(t))$$

$\gamma = g \cdot \phi_v(t)$ integral curve of X_v through $g \in G$

$$\dot{\gamma}(t) = \frac{d}{ds} \Big|_{s=0} \gamma(t+s) = \frac{d}{ds} \Big|_{s=0} g \cdot \phi_v(t+s)$$

$$= d(\lambda_{g \cdot \phi_v(t)}) \cdot \underbrace{\frac{d}{ds} \Big|_{s=0} \phi_v(s)}_v = d(\lambda_{g \cdot \phi_v(t)}) \cdot v = X_v(\gamma(t))$$

ii) $\phi_v(t) = \phi_{vf}(t)$ (both integral curves of same vf)

Define: $\exp: \mathfrak{g} \rightarrow G, v \mapsto \phi_v(1)$

i) $\exp(t \cdot v) \exp(s \cdot v) = \exp((t+s) \cdot v)$

ii) $\frac{d(\exp)_0}{\text{id}_{\mathfrak{g}}} : \mathfrak{g} \rightarrow \mathfrak{g}, v \mapsto v$

iii) $\phi: G \rightarrow H \Rightarrow \phi(\exp(v)) = \exp(d\phi \cdot v)$

Matrix groups

$$\exp = \text{Exp}: \text{Mat}(n, \mathbb{R}) \rightarrow \text{GL}(n), v \mapsto \sum_{i=0}^{\infty} \frac{v^i}{i!} \quad \text{matrix multiplication}$$

Baker-Campbell-Hausdorff formula

$$\exp(v) \exp(w) = \exp\left(v + w + \frac{1}{2}[v, w] + \frac{1}{12}([v, [v, w]] - [w, [w, v]]) + \dots\right)$$

The Lorentz group

Minkowski space: $(\mathbb{R}^{1,3}, \eta = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix})$

$$\mathcal{L} = O(1,3) = \{ \Lambda \in \text{Mat}(4,4) \mid \Lambda^T \eta \Lambda = \eta \}$$

$$\Lambda = \begin{pmatrix} \Lambda_{00} & \lambda^T \\ \lambda_1 & \hat{\Lambda} \end{pmatrix}, \quad \Lambda_{00} \in \mathbb{R}; \lambda_1, \lambda_2 \in \mathbb{R}^3; \hat{\Lambda} \in \text{Mat}(3,3)$$

$$\Lambda^0 \eta \Lambda = \begin{pmatrix} \Lambda_{00}^2 - \lambda_1^T \lambda_1 & \Lambda_{00} \lambda_2^T - \lambda_1^T \hat{\Lambda} \\ \Lambda_{00} \lambda_2 - \hat{\Lambda}^T \lambda_1 & -\hat{\Lambda}^T \hat{\Lambda} + \lambda_2 \lambda_2^T \end{pmatrix} = \begin{pmatrix} 1 & \\ & \mathbb{I}_3 \end{pmatrix} \quad (*)$$

Polar decomposition of $O(1,3)$

Note that $\eta^t = \eta$, $\eta^2 = 1_4 \Rightarrow \eta \in O(1,3)$, $\Lambda^t = \eta \Lambda^t \eta \in O(1,3)$

$\Rightarrow \Lambda^t \Lambda \in O(1,3)$ is symmetric, > 0

$$\Lambda := \sqrt{\Lambda^t \Lambda}$$

$$\Lambda^t = \Lambda, \Lambda > 0, \Lambda \in O(1,3)$$

$$R := \Lambda \Lambda^{-1} \in O(1,3)$$

$$R^t R = (\Lambda^{-1})^t \Lambda^t \Lambda \Lambda^{-1} = 1_4 \Rightarrow R \in O(1,3) \cap O(4)$$

$$\Rightarrow R \in O(1) \times O(3), R = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & & & \\ 0 & & Q & \\ 0 & & & \end{pmatrix}, \epsilon \in O(1) = \{\pm 1\}, Q \in O(3)$$

$$\Lambda = \begin{pmatrix} \epsilon & 0 \\ 0 & Q \end{pmatrix} \Lambda, \Lambda = \begin{pmatrix} \lambda_{00} & \lambda_{0i} \\ \lambda_i & \lambda_{ij} \end{pmatrix}, \Lambda^t = \Lambda$$

$$[x] \Rightarrow \lambda_{00} = \sqrt{1 + \lambda^t \lambda}, \Lambda = \sqrt{1 + \lambda \lambda^t}$$

$$\Lambda = \begin{pmatrix} \epsilon & 0 \\ 0 & Q \end{pmatrix} \begin{matrix} \leftarrow \text{time-reversal} \\ \leftarrow \text{space transf.} \end{matrix} \cdot \begin{pmatrix} \sqrt{1 + \lambda^t \lambda} & \lambda^t \\ \lambda & \sqrt{1 + \lambda \lambda^t} \end{pmatrix} \leftarrow \text{boosts}$$

$$O(1,3) \stackrel{\text{homom.}}{\cong} O(1) \times O(3) \times \mathbb{R}^3$$

$\rightarrow$ 4 components distinguished by

$$\det \Lambda = \epsilon \det(Q) = \pm 1 \quad (+1: \text{proper, preserve orientation})$$

$$\text{sign}(\lambda_{00}) = \epsilon = \pm 1 \quad (+1: \text{orthochronous, time-direction preserved})$$

Connected component of $e \in O(1,3)$ is called $O(1,3)_0$ & $O(1,3)$

$$O(1,3)/O(1,3)_0 = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{1, P, T, PT\}, \quad P = -1_4, \text{ parity-reversal}$$

$$T = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \text{ time-reversal}$$

Lie algebra $O(1,3)$

$$O(1,3) \text{ is level set: } f(\Lambda) = \Lambda^t \eta \Lambda - \eta$$

$$df_v V = V^t \eta + \eta V$$

$$\ker(df_v) = \{V \mid V^t = -\eta V \eta\}$$

$$T_e O(1,3) = O(1,3)$$

$$V = \begin{pmatrix} V_0 & c^t \\ d & \vec{v} \end{pmatrix} = \begin{pmatrix} -V_0 + d^t \\ +c & -\vec{v}^t \end{pmatrix}$$

$$\Rightarrow v_1 = 0, c = d, \hat{v}^F = -\hat{v}$$

$$v = \begin{pmatrix} 0 & c^t \\ c & 0 \end{pmatrix}$$

$$\text{basis: } \omega_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \omega_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \omega_3 = \begin{pmatrix} 0 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$K_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, K_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$\mathfrak{o}(1,3) = \text{span}\{\omega_i, K_i\} \leftarrow 6 \text{ dimensional real Lie algebra}$

$$[\omega_i, \omega_j] = \sum_k \epsilon_{ijk} \omega_k$$

$$[K_i, K_j] = -\sum_k \epsilon_{ijk} \omega_k$$

$$[\omega_i, K_j] = \sum_k \epsilon_{ijk} K_k$$

$$SL_2(\mathbb{C}) = \{A \in \text{Mat}(2, 2, \mathbb{C}) \mid \det(A) = 1\}$$

$$sl_2(\mathbb{C}) = \{a \in \text{Mat}(2, 2, \mathbb{C}) \mid \text{tr}(a) = 0\} = \text{span}_{\mathbb{C}}\{j_1 = \frac{1}{2} \sigma_1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$j_2 = \frac{1}{2} \sigma_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, j_3 = \frac{1}{2} \sigma_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\} \quad \leftarrow \sigma_i \text{ Pauli matrices}$$

$$[j_a, j_b] = \sum_c \epsilon_{abc} j_c$$

$\rightarrow$ complex 3-dimensional Lie algebra

Choose a real structure

$$sl_2(\mathbb{C}) = \text{span}_{\mathbb{R}}\{j_1, j_2, j_3\} \otimes_{\mathbb{R}} \mathbb{C}$$

$$\varphi: (sl_2(\mathbb{C}))_{\mathbb{R}} \rightarrow \mathfrak{o}(1,3), j_a \mapsto \omega_a, -ij_a \mapsto K_a$$

isomorphism of real Lie algebras

$sl_2(\mathbb{C})$ is in fact simply connected!

$\Rightarrow \varphi$ integrates to a unique

$$\Phi: SL(2, \mathbb{C}) \rightarrow \mathfrak{o}(1,3)_0 \text{ which is possibly a covering}$$

$$d\Phi_e = \varphi$$

identity: $\mathbb{R}^{1,3} \xrightarrow{\cong} iU(2)$, $x \mapsto \mathcal{Z}(x) = \sum_i x_i \mathcal{Z}_i = \begin{pmatrix} x_0 + x_3 & x_1 + ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}$

hermitian 2×2 -matrices

$$x^\dagger \eta x = \det(\mathcal{Z}(x))$$

Now $SL(2, \mathbb{C}) \curvearrowright iU(2)$, $\mathcal{Z}(x) \mapsto A \mathcal{Z}(x) A^\dagger$, $A \in SL(2, \mathbb{C})$

But this action preserves η :

$$\det(A \mathcal{Z}(x) A^\dagger) = \det(\mathcal{Z}(x)) \quad \text{since } \det(A) = 1$$

$$\Rightarrow A \mathcal{Z}(x) A^\dagger = \mathcal{Z}(\psi(A) \cdot x), \quad \text{where } \psi: SL(2, \mathbb{C}) \rightarrow O(1,3)$$

$$\text{check: } d\psi_e = \varphi \Rightarrow \psi = \varphi$$

$A(t)$ path in $SL(2, \mathbb{C})$: $A(0) = 1$, $\dot{A}(0) = a$

$$\left. \frac{d}{dt} \right|_{t=0} A(t) \mathcal{Z}(x) A^\dagger(t) = a \mathcal{Z}(x) + \mathcal{Z}(x) a^\dagger = \mathcal{Z}(d\psi_e a) \cdot x$$

$$a = j_i = \frac{i}{2} \mathcal{Z}_i \Rightarrow \left. \frac{d}{dt} \right|_{t=0} A(t) \mathcal{Z}(x) A^\dagger(t) = \frac{i}{2} [\mathcal{Z}_i, \mathcal{Z}(x)] = \sum \mathcal{Z}_c(w_i \cdot x)_c = \mathcal{Z}(w_i \cdot x)$$

$$a = -j_i \Rightarrow \left. \frac{d}{dt} \right|_{t=0} A(t) \mathcal{Z}(x) A^\dagger(t) = \mathcal{Z}(v_i \cdot x)$$

$$\Rightarrow \varphi = d\psi_e: j_i \mapsto w_i, -j_i \mapsto v_i$$

$$\Rightarrow \psi = \varphi$$

$$\ker(\varphi) = \{A \in SL(2, \mathbb{C}) \mid A \mathcal{Z} A^\dagger = \mathcal{Z} \forall \mathcal{Z}\} = \{\pm 1, \pm \mathbb{1}_2\}$$

Remark: complexification

$$o(1,3)_\mathbb{C} = o(1,3) \otimes_\mathbb{R} \mathbb{C} \rightarrow 6\text{-dimensional, complex Lie algebra}$$

$$\text{basis: } J_i^\pm := \frac{1}{2}(\omega_i \pm i\kappa_i)$$

complex basis of $o(1,3)_\mathbb{C}$

$$[J_i^+, J_j^-] = 0 \quad \forall i, j$$

$$[J_i^\pm, J_j^\pm] = \sum_k \varepsilon_{ijk} J_k^\pm$$

$$\mathfrak{o}(1,3)_{\mathbb{C}} \xrightarrow{\cong} \mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_2(\mathbb{C})$$

$$J_i^+ \mapsto (j_i, 0)$$

$$J_i^- \mapsto (0, j_i)$$

isomorphism of complex Lie algebras

$$\boxed{\mathfrak{o}(1,3)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_2(\mathbb{C})}$$